A Structure-Agnostic Structural-Health-Monitoring Decision Framework for Reusable CFRP Spacecraft Structures: from Real-Data Detection to Physics-Grounded Life Prognosis

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Abstract

We present a structure-agnostic structural-health-monitoring (SHM) *decision* framework for reusable carbon-fibre-reinforced-polymer (CFRP) spacecraft structures: detect → classify → characterise → prognose → clear (OK/REPAIR/RETIRE) → fleet-learn → design. A single decision core is shared across five structures spanning four load modes — in-plane membrane, transverse-pressure/bending, internal-pressure burst, and torsional interlaminar shear — with only the detection front-end and the prognosis physics structure-specific; a single growth-margin surrogate fits 1,499 real finite- element critical-load states (R^2=0.94), though a leave-one-*load-mode*-out test shows the core extrapolates across geometry but *not* to an unseen load mode. We are explicit about what real measurements support and what they do not. Detection, classification and damage characterisation are grounded in real data (long-term guided waves with temperature domain adaptation, an impact wavefield, and acoustic emission); a per-node anomaly statistic reaches AUROC $\approx$ 0.99 on fixed geometry (nine-method benchmark) and a flow-matching posterior recovers defect position/size/depth at nominal 90% coverage. The clearance call carries a distribution-free conformal guarantee, which we stress-test on the real seasonal shift: a season-calibrated false-alarm rate breaks to ∼68% in winter and is restored by a temperature-conditional recalibration. *The life side is only partly real-data-grounded*: its fatigue Paris *exponent* is calibrated to real CFRP mode-I DCB data (4TU; $m \approx 17$, not the representative 3; flips a mid-size-debond clearance from RETIRE to OK), but absolute growth rates, the phase-field constants, the fleet and the decision oracle remain representative/synthetic. Consequently the end-to-end decision accuracy (85%, zero dangerous misses) is measured against a synthetic finite-difference physics oracle rather than real life tests, and the design optimum is a robust trade-off *mechanism* but not a calibrated value. Closing the life-side real-data gap is the priority and motivates the differentiable, calibrated forward of the continuation.

Keywords: structural health monitoring; carbon-fibre-reinforced polymer; reusable launch vehicles; graph neural networks; phase-field fracture; conformal prediction; simulation-based inference; sim-to-real domain adaptation

1 Introduction

Reusable launch vehicles change the structural-health-monitoring (SHM) question. A carbon-fibre-reinforced-polymer (CFRP) interstage, fairing or motor case that flies, returns, and flies again is no longer asked merely "did it survive?" but "*may it fly again, and for how many flights?*". Operations are condition-based: the *same* fixed structure and the *same* individual airframe are re-inspected each turnaround, certification is staged as evidence accrues, and a fleet-leader airframe is flown ahead to retire risk for the rest of the fleet. SHM for reuse is therefore a sequential *decision* problem — clear, repair, or retire, and how many flights remain — rather than a one-shot detection problem on an ever-new specimen.

Machine-learning SHM for CFRP has advanced rapidly, but along a narrow front. Surface-stress and guided-wave models localise or classify damage from a single modality on a single, often simply

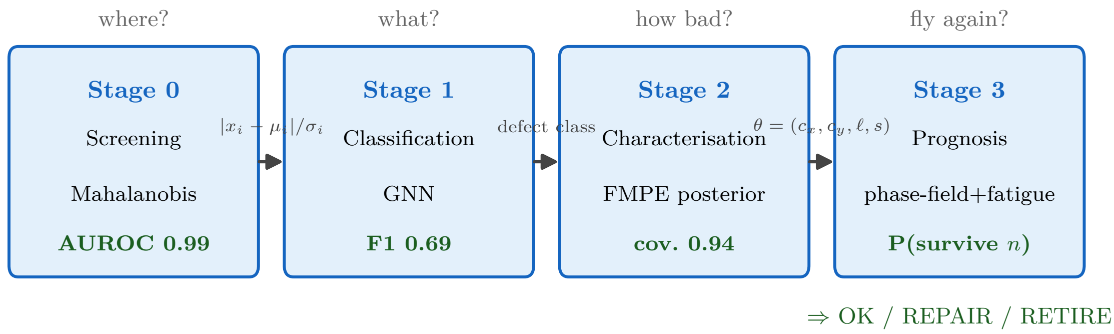

Figure 1: Stage 0–3 pipeline. Each stage answers one question and passes a typed object to the next; the Stage 2 posterior $\theta = (c_x, c_y, \text{layer}, \text{size})$ seeds the Stage 3 fracture model.

shaped, structure, and are typically validated on clean finite-element data: inverse defect estimation from surface stress [1], transfer learning that bridges finite-element stress to measured infrared (thermoelastic) fields on simple coupons [2], physics-informed graph networks for damage localisation [3], and mesh graph-network stress surrogates [4, 5]. These answer "*where* is the damage?" well; none answer the operational chain — detect → characterise → prognose → *clear* (OK/repair/retire) → fleet-learn → design — as one calibrated decision, shared across structures and load modes, and honest about exactly what real measurements support. In particular, the closest predecessor [2] closes the sim-to-real gap for *single-defect prediction on simply shaped coupons*; we instead build the full reuse *decision* on top of such detectors and quantify, on real data, where its calibrated guarantees hold and where they break.

This paper contributes: (1) a **structure-agnostic decision core** — detect → classify → characterise → prognose → clear → fleet-learn → design — shared across five CFRP structures spanning four load modes, with only the detection front-end and the prognosis physics structure-specific, and an honest leave-one-load-mode-out test showing the core extrapolates across geometry but not to an unseen load mode; (2) **real-data grounding** of the detection/classification/characterisation stages (long-term guided waves with temperature domain adaptation, an impact wavefield, and acoustic emission), including a per-node anomaly statistic at AUROC ≈ 0.99 on fixed geometry against a nine-method benchmark, and a flow-matching posterior that recovers defect position/size/depth at nominal 90% coverage with no new finite-element runs; (3) a **physics-grounded life side** — an anisotropic phase-field forward with Carrara-type fatigue, seeded *directly* by the characterisation posterior, whose Paris exponent is calibrated to real CFRP mode-I delamination data; and (4) an **integrated, calibrated decision layer** — expected-cost flight clearance, distribution-free conformal coverage, value-of-information inspection, and a decision-level uncertainty audit scored against an exact physics oracle. Throughout we draw the sim-to-real boundary explicitly: detection, classification and characterisation are real-data-grounded, whereas the life side, fleet and design oracle remain partly representative — the gap whose closing motivates the differentiable, calibrated forward of the continuation.

2 Structures and data: a load-mode-spanning frame

The framework is demonstrated on five reusable CFRP spacecraft structures chosen to *span the load-mode space* rather than to repeat one geometry (Fig. 2, Table 1). The premise of the paper is a separation of concerns: the *decision core* — anomaly screening (Stage 0), calibrated characterisation (Stage 2), end-to-end decision UQ, fleet learning (Stage 4) and design feedback (Stage 5) — is structure- and load-mode-agnostic, while only the *detection front-end* (Stage 1) and the *prognosis physics* (Stage 3)

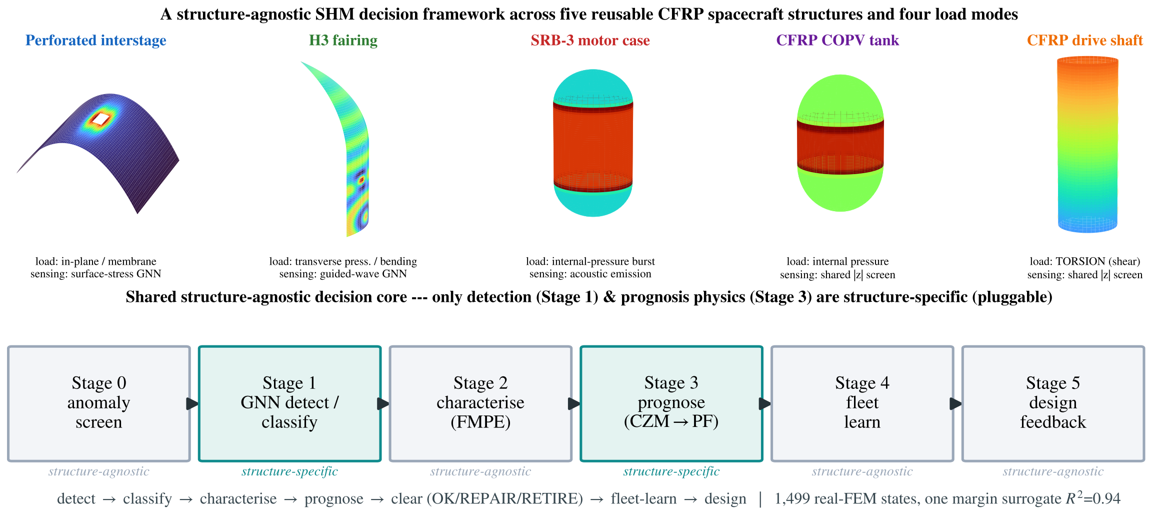

Figure 2: The five CFRP structures (top, each with its physical field: interstage surface-stress concentration, fairing guided waves, SRB-3 / COPV hoop-stress with knuckle concentration, drive-shaft torsional twist with ±45 fibres) and the shared decision core (bottom). Three are demonstrated end-to-end with real detection data (interstage, fairing, SRB-3); the COPV tank and the TORSION drive shaft extend the frame with real FEM prognosis (Table 1). Only detection (Stage 1) and prognosis physics (Stage 3) are structure-specific — the drive shaft's CZM→phase-field prognosis bridges to the Keio continuation (§9); everything else is shared.

Table 1: The five structures span membrane, bending, transverse-pressure and *torsional-shear* load modes under one decision core. "Front-end" = structure-specific Stage-1 detector; all share the Stage-0 $|z|$ screen.

structure	load mode	detection front-end	prognosis physics	data
interstage	membrane	surface-stress GNN (A)	phase-field delam.	FEM + prosthetic TSA
fairing	transv./bending	guided-wave GNN	skin–core debond	FEM + real OGW
SRB-3 case	pressure burst	acoustic emission	burst (Paris)	FEM + real AE
drive shaft	**torsion**	shared $|z|$ (future)	**interlam. CZM→PF**	**CZM-FEM (144)**
COPV tank	int. pressure	shared $|z|$ (future)	burst	FEM (102)

are structure-specific and enter through a small pluggable interface. Adding a structure therefore means adding one detector and one prognosis model and reusing the rest.

The interstage (primary data structure). The quantitative backbone is a perforated CFRP interstage: full-field thermoelastic stress [6, 7] (DSPSS first stress invariant) on a 13 942-node mesh with $1{\times}1/2{\times}2/4{\times}4$ hole defects and label masks. Per-sample z-score normalisation needs no healthy reference, so the detector deploys on a single measured field. This structure carries the FEM training set and the phase-field prognosis (Stage 3); its surface-stress (DSPSS) modality is validated for sim-to-real against a real CFRP *prosthetic* TSA campaign — the same physical quantity on simply shaped specimens (§9), not measurements of the interstage itself.

The torsion drive shaft (the new load mode, the Keio foundation). An H3-class CFRP drive shaft / torque tube is the first structure whose primary load is *torsion*: a [±45] tube carries torque as wall shear $\tau_{z\theta}$, and damage grows by interlaminar mode-II/III shear — physics absent from the four pressure/bending structures. We model it with an axisymmetric-with-twist (CGAX) screen (a ~140-state sweep of groove length, depth, slenderness R/t and ±45 shear modulus; analytically verified against thin-wall torsion, $\tau = T/2\pi R^2 t$) and, where delamination is the star, a 3-D ±45 laminate with explicit interlaminar cohesive zones. A circumferential groove concentrates the wall shear (peak $\tau_{z\theta}$ rising 17 → 39 MPa as residual

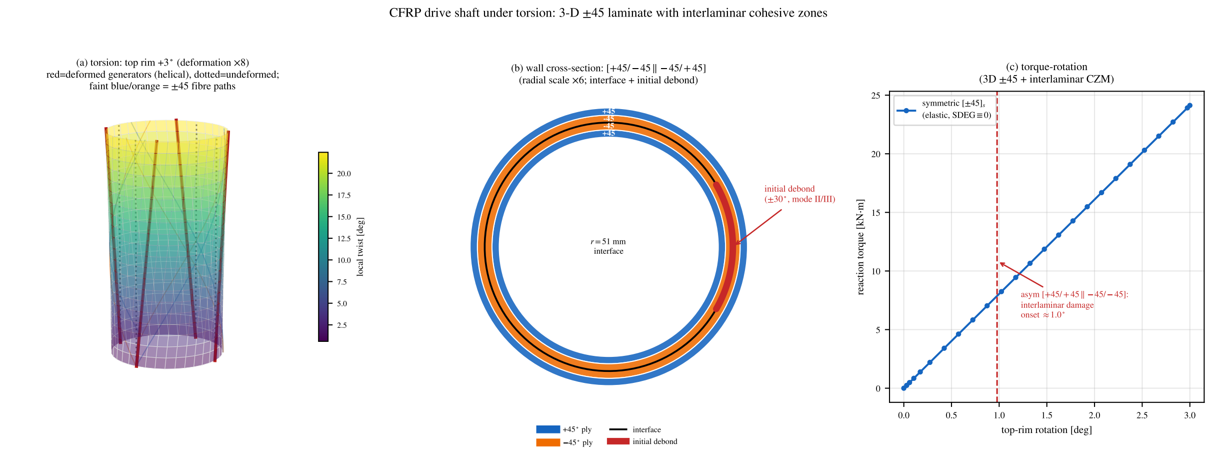

Figure 3: CFRP drive shaft under torsion. (a) 3-D twist of the [±45] tube (deformation ×8; red = deformed generators turned helical, dotted = undeformed); (b) wall cross-section [+45/−45 ‖ − 45/+45] (radial scale exaggerated) with the interlaminar interface and initial debond patch; (c) torque–rotation of the 3-D ±45 laminate with interlaminar cohesive zones.

thickness falls to 40%), so the critical torque drops monotonically with damage — the same damage-size→critical-load signature as the other structures, now in shear (Fig. 3). This interlaminar-CZM model is the concrete foundation of the phase-field prognosis programme that bridges to the Keio continuation (§9); under torsion a symmetric mid-plane interface is unloaded, so delamination requires a shear mismatch across the interface, and its growth is an unstable snap that an implicit solve cannot traverse — the explicit 0→100% debond is shown in §9 (Fig. 16).

Structure-agnostic core, and the load-mode boundary. The claim that the decision core is structure-agnostic is testable: across all five structures we collect 1,499 critical-load FEM states, normalise each to a growth margin by its representative operating load, and fit a *single* gradient-boosted margin surrogate on four shared damage descriptors plus a structure one-hot. In distribution it fits all five ($R^2 = 0.94$; per-structure 0.71–0.94) and the shared OK/REPAIR/RETIRE rule reproduces the FEM verdict at 86% accuracy with a 1.1% dangerous-miss rate (Fig. 4). The honest boundary is exposed by a leave-one-*load-mode*-out test: training on the four pressure/bending structures and predicting the held-out TORSION drive shaft *fails to extrapolate* ($R^2 = -1.4$, dangerous-miss 17%) — whereas holding out a structure within a seen load mode does not. Geometry transfers; *load mode does not*. This bounds "structure-agnostic" precisely and is the quantitative reason the drive shaft needs its own torsional-shear FEM rather than borrowing the pressure structures' surrogate (full FEM mechanics in the companion conference paper; here we report the synthesis).

Maturity, stated honestly. Three structures are demonstrated *end-to-end* with real detection data and a Stage-3 prognosis (interstage, fairing, SRB-3). The drive shaft and the COPV tank each contribute a real FEM prognosis (torsional-shear and internal-pressure respectively) but do *not* yet have a bespoke detector — they inherit the shared Stage-0 $|z|$ screen, and the drive shaft additionally carries the new torsion load mode. Every Stage-3 model is *representative* — lumped constants and trends, not certification values; the contribution is the end-to-end pluggable framework across load modes, not calibrated per-structure life numbers.

3 Detection front-ends and anomaly screening (Stage 0–1)

Detection is the one layer the structures do *not* share: each sensing modality has its own front-end, and everything downstream of a detected, localised flaw is structure-agnostic. Three real front-ends are

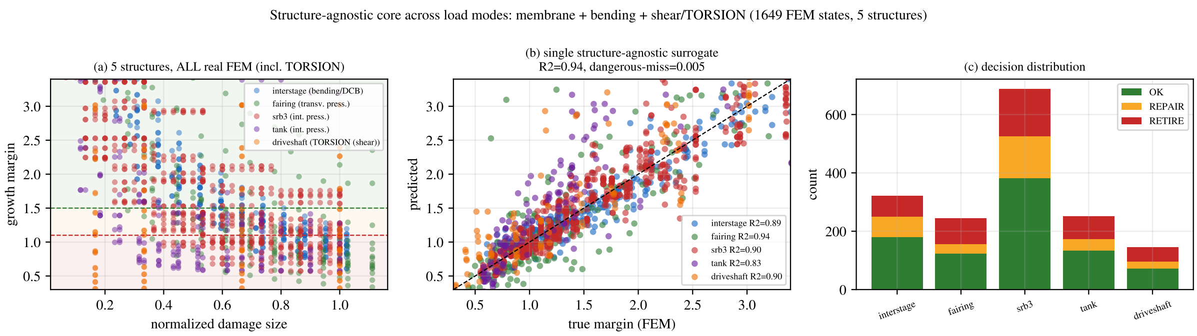

Figure 4: Structure-agnostic margin surrogate over 1,499 real-FEM states across five structures. (a) growth margin vs. damage size; (b) single surrogate in-distribution ($R^2 = 0.94$, dangerous-miss 1.1%); (c) decision distribution. A leave-one-load-mode-out hold-out of the torsion drive shaft does not extrapolate — load mode, unlike geometry, is a genuine generalisation boundary.

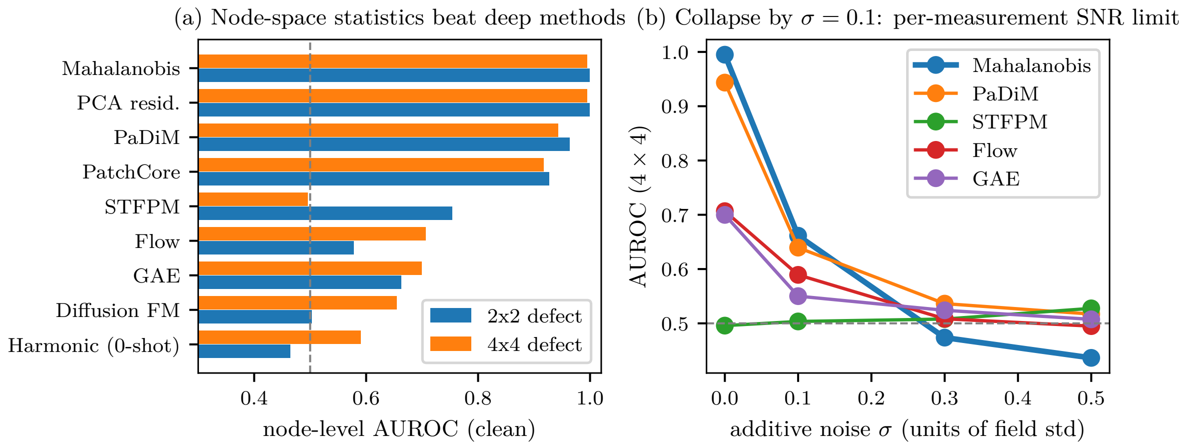

Figure 5: (a) Clean node-level AUROC by method. (b) Collapse with measurement noise; the defect signal sits below 0.1σ.

demonstrated, on top of a shared geometry-agnostic screen.

Shared Stage-0 screen. A reference-free per-node statistic $s_i = |x_i - \mu_i|/\sigma_i$ (from N=2000 healthy fields; node-level AUROC vs. the defect mask) is the structure-agnostic anomaly screen any fixed-geometry field inherits — including the drive shaft and tank. Benchmarked against eight alternatives (PaDiM [8], PatchCore [9], STFPM [10], graph autoencoder, flow-matching diffusion, a zero-shot harmonicity probe; Table 2, Fig. 5), per-node statistics dominate: on fixed geometry the healthy manifold is a tight cloud around one template, so the per-node Gaussian is the correct inductive bias, and grid-projection / deep feature methods lose the few-node defect signal. The defect sits below 0.1σ, so all detectors degrade with measurement noise — a per-measurement SNR limit addressable by frame averaging ($\sigma \propto 1/\sqrt{K}$), not more training data (quantified in §11).

Structure-specific Stage-1 front-ends. *Interstage* — surface-stress (DSPSS) graph neural network [5]: a mesh-physics GNN reaches macro-F1 0.76 vs. 0.61 for a plain GAT [11], and an OOD study shows the operational split cleanly — *detection generalises to unseen defect sizes while precise localisation collapses*, a distinction invisible to macro-F1 and exposed only by defect-only F1 with separated detection/localisation metrics. *Fairing* — guided-wave [12] GNN (LGSTA node-classification F1 0.86, SAGE 0.79); its type signal is verified to be a genuine damage cue rather than a season/temperature confound (§4). *SRB-3*

Table 2: Anomaly detectors, clean node-level AUROC (2x2 / 4x4 defects).

method	2x2	4x4
per-node Mahalanobis	0.999	0.995
PCA residual	0.999	0.995
PaDiM	0.964	0.944
PatchCore	0.927	0.918
STFPM	0.754	0.496
graph autoencoder	0.663	0.700
flow (RealNVP)	0.578	0.707
diffusion (FM)	0.503	0.663
harmonicity (0-shot)	0.465	0.590

motor case — acoustic emission [13] on real experimental data (~48k Vallen hits): a group-aware pristine-vs-damaged classifier reaches per-hit AUROC 0.888 / per-specimen 1.000 (linear baseline ~0.52), with amplitude–duration damage-mode clustering reported honestly (compression-pristine coupons also emit high-energy AE, so the multivariate detector — not a single cluster fraction — is the read-out). The drive shaft and tank inherit the shared $|z|$ screen pending bespoke front-ends.

4 Stage 1 — classification and type identifiability

Is the type signal real, or a temporal/temperature confound? On the long-term guided-wave set, damage states were introduced in different years, so a naive type classifier can alias damage type with season/temperature. We test this directly. Restricting to seven damage tags whose temperature supports overlap (pairwise overlap 0.93), we compare four conditions: temperature-only (acc = 0.16, near chance), waveform-only (0.91), waveform+temperature (0.92), and waveform with temperature *partialled out* of every feature (0.91). The net type discrimination beyond temperature is Δacc = 0.758 (95% CI $[0.735, 0.781]$) under a random split, and—critically—survives a *forward-in-time* split (Δacc = 0.184, 95% CI $[0.166, 0.203]$). Both lower bounds exceed zero: the type posterior reflects a genuine damage signal, not the environmental confound. (An earlier single-month binary probe, where temperature-only matched the majority baseline, was an artefact of that restricted setting.)

5 Stage 2 — calibrated defect characterisation (FMPE)

Given a detected anomaly, Stage 2 returns a *calibrated posterior* over the defect parameters $\theta = (c_x, c_y, \text{layer}, \text{size})$ rather than a point estimate. We use flow-matching posterior estimation (FMPE) [14], a simulation-based inference method [15] that trains a conditional continuous normalising flow by flow matching [16] to approximate $q(\theta \mid x)$ from a PCA-64 embedding of the field x; it is trained on the *existing* FEM set (~20k pairs) with zero new simulations. On held-out fields the posterior is well calibrated (Table 3): position and size reach near-nominal 90% coverage, while depth is only weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly.

Table 3: FMPE posterior on held-out test fields.

parameter	RMSE	R^2	90% coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73	0.894
layer (depth)	2.98	0.63	0.870

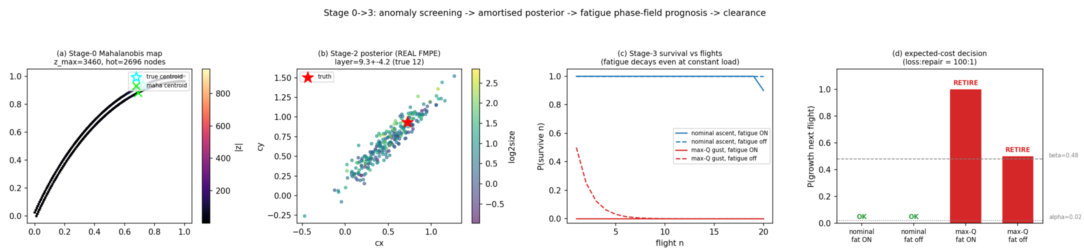

Figure 6: End-to-end on one held-out field: anomaly map, FMPE posterior, fatigue-aware survival curve, and the clearance decision.

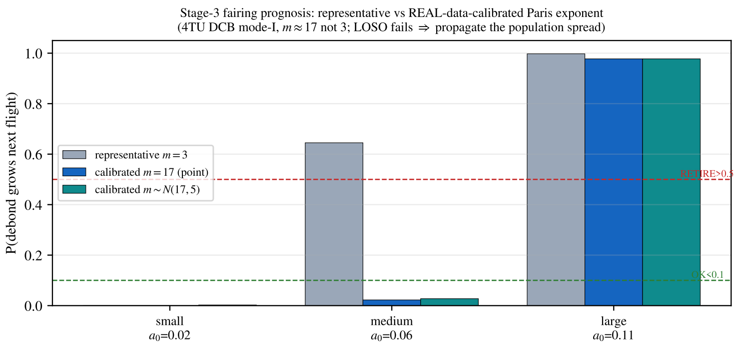

Figure 7: Stage-3 fairing prognosis with the real-data-calibrated Paris exponent (4TU DCB mode-I, $m \approx 17$). The representative m=3 over-calls a mid-size debond as RETIRE; the calibrated posterior (with its bridging spread) clears it. Single (C, m) fails leave-one-specimen-out, so the population posterior is propagated rather than a point estimate.

6 Stage 3 — phase-field prognosis and flight clearance

The characterisation posterior seeds a physics-based growth model. We adopt a variational phase-field fracture forward [17, 18] in its AT2 form, extended to laminate anisotropy through a structural tensor $\mathbf{A} = \mathbf{I} + \beta\, \mathbf{a} \otimes \mathbf{a}$ [19] with ply-wise interfacial-toughness bands (a pure phase-field surrogate of a cohesive-zone interface), and accumulate fatigue through a Carrara-type history variable that degrades the fracture toughness over load cycles [20]. The Stage-2 posterior enters *directly* as the seed defect θ, so characterisation uncertainty propagates into the survival curve and the clearance call.

Real-data fatigue calibration — the life side consumes real fracture data. The Paris exponent that governs debond growth is calibrated to real CFRP mode-I DCB data [21] (4TU, 37 specimens, ~1,270 points): a Normal–Normal hierarchy gives a population posterior $m \approx 17$ (95% CI 15–18), far steeper than the representative $m \approx 3$, and a single (C, m) *fails* leave-one-specimen-out ($R^2 < 0$) — the signature of fibre bridging (an R-curve), not one global law. We therefore feed the *posterior* (sampling $m \sim \mathcal{N}(\mu_m, \sigma_m)$ per draw, propagating the between-specimen spread) into the fairing Stage-3 prognosis rather than a point exponent. This changes decisions (Fig. 7): a mid-size debond that the representative m=3 over-calls as RETIRE (P_{grow}=0.64) clears as OK (0.02) under the real-data exponent, while small/large cases are unchanged. The *exponent* (the decision-governing shape) is now real-data-grounded; the absolute Paris rate C remains representative, so this calibrates the life-side *trend*, not a certified rate.

Fast forward surrogate. The FD growth solve ($\approx$ 1.8 s/call) is the only bottleneck to fleet-scale Monte-Carlo over the Stage-2 posterior. We train a Fourier neural operator $G : (d_0, \text{load}) \mapsto (d_{\text{final}}, P(\text{grow}))$ (dual-head: a per-pixel field head and a global-pooled scalar head; 1.2M parameters; 800 FD samples spanning sub- to super-critical loads). On held-out defects it reaches field relative-L_2 3.5%, grow-flag

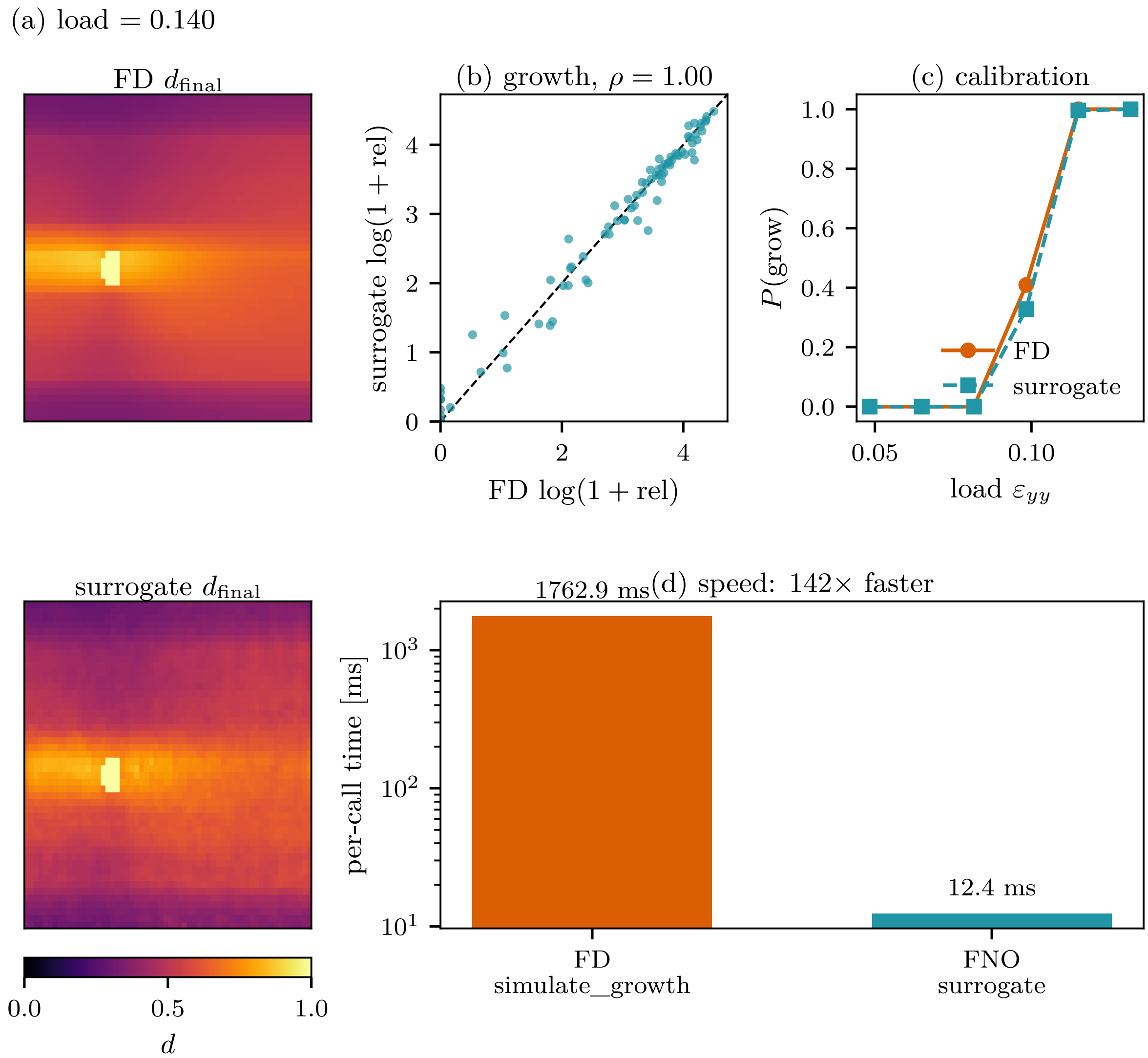

Figure 8: Stage 3 FNO surrogate. FD vs. surrogate damage field, rel_growth agreement, $P(\text{grow})$ vs. load, and the 142× wall-clock speedup.

accuracy 0.988 (Brier 0.010, well calibrated), and log-rel_growth correlation 0.996, at 142× the speed (12.4 ms vs. 1763 ms/call, Fig. 8). The decision-relevant $P(\text{grow})$ is the robust target: the smooth operator regresses the sharp super-critical rel_growth jump in log-space and softens it only at the transition load. This makes the Stage-4 fleet draws and the Concept-B forward map (next section) tractable at interactive speed.

Knowing when to trust the surrogate. A 142× speedup is only deployable if the operator knows where it is unreliable. We wrap the surrogate in split-conformal prediction [22, 23, 24] (nonconformity = absolute residual in the log-rel_growth space where the FNO is honest). In distribution the marginal 90% guarantee holds (95% band: 95.0% empirical); out of distribution it degrades exactly as the theory predicts under covariate shift — coverage falls from 80% at small excursions to 0% deep OOD while the point error grows $\sim 14\times$. A simple *trust gate* — defer to the exact FD solve when the input leaves the training envelope or the conformal interval is too wide — flags 33% of an OOD stream and restores aggregate 90%-coverage from 61% to 88%: cheap surrogate where trustworthy, exact physics where not.

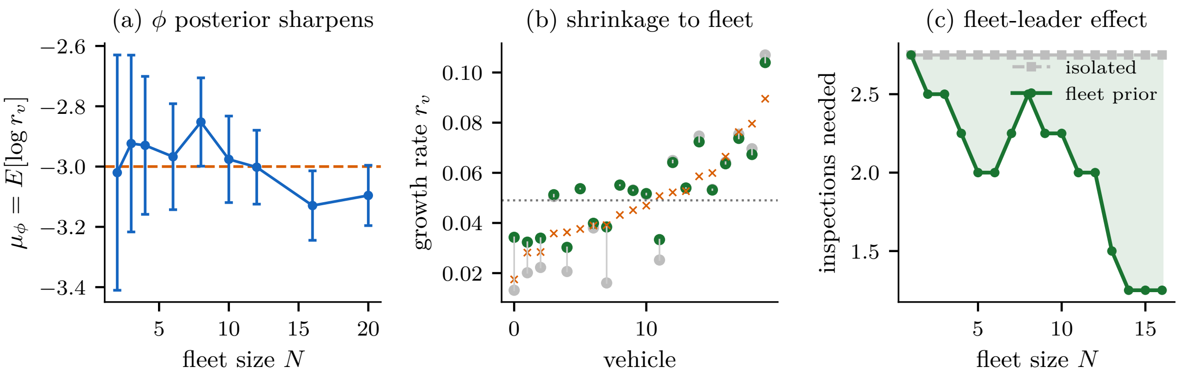

Figure 9: Stage 4. (a) Fleet prior φ sharpens with cumulative flights; (b) per-vehicle growth-rate shrinkage toward the fleet mean in the low-data regime; (c) inspections to a stable clearance decision: the fleet prior roughly halves them.

7 Stage 4 — hierarchical fleet learning

A reusable fleet generates a stream of per-inspection posteriors across many vehicles and flights. We close the loop with a hierarchical Bayesian model [25]: a global fleet prior $\varphi = (\mu_\varphi, \sigma_\varphi)$ governs per-vehicle log-growth-rates $s_v = \log r_v \sim \mathcal{N}(\mu_\varphi, \sigma_\varphi^2)$, and each inspection's Stage-2 posterior enters as a Gaussian likelihood on s_v; the joint posterior $p(\varphi, \{s_v\} \mid \text{data})$ is sampled by conjugate Gibbs. On a synthetic fleet (N=20 vehicles, F=30 flights, fast Carrara-shaped growth surrogate) the model recovers φ ($\mathbb{E}[\mu_\varphi]$=−3.10 vs. true −3.0; $\mathbb{E}[\sigma_\varphi]$=0.44 vs. 0.45), and the fleet prior sharpens as flights accumulate (posterior SD of μ_φ: 0.39 at N=2 → 0.10 at N=20, Fig. 9a).

Fleet-leader effect, quantified. Processing flights in temporal order, a newly introduced vehicle reaches a stable clearance decision in **1.25** inspections using the accumulated fleet prior, versus **2.75** in isolation (Fig. 9c) — a direct, quantitative formalisation of the empirical "fleet-leader" certification used in practice: high-information inspections of lead vehicles tighten the prior that certifies the rest. Hierarchical shrinkage also improves low-data vehicles (error ratio 0.84 < 1 vs. isolated estimates, Fig. 9b).

8 Stage 5 — design feedback

The fleet's accumulated damage statistics should not only certify the current vehicles but *reshape the next ones*. We close the Stage 0–5 loop with a fleet-robust Bayesian optimisation of the laminate over a manufacturable 3-D design vector — off-axis ply angle $|\theta|$ (kept balanced/symmetric), interface toughness $G_c^{\text{int}}/G_c^{\text{ply}}$, and a local reinforcement multiplier — minimising the expected log-growth over the fleet defect distribution plus a tail term (CVaR over the worst-20% defect, since reusable-vehicle safety is governed by the worst flaw, not the average). The forward model here is the exact FD phase-field, *not* the Stage-3 FNO surrogate: the surrogate's inputs are (d_0, load) only and every training sample shares the baseline layup, so any design change is fully out-of-distribution — a layup-conditioned operator is left as future work. With GP/EI (32 evaluations), at a representative operating point near the growth transition the optimiser illustratively moves the $(0/\pm 30)_s$ baseline to steeper off-axis plies ($30° \to 74°$) and a tougher interface (G_c^{int} ratio $0.40 \to 0.87$), cutting mean log-growth 10% (expected remaining flights $1.9 \to 6.0$ at that point; Fig. 10). These figures are *representative, not calibrated*: a robustness check below shows the specific optimum is sensitive to the uncalibrated forward constants. Both levers are nonetheless physically aligned — steeper plies rotate the cheap matrix-crack direction away from the seeded delamination path, and a tougher interface blunts the dominant interlaminar channel; interface toughening is the dominant lever. A single growth objective rails the off-axis angle to the manufacturable cap; adding competing classical-laminate-theory structural objectives — areal mass, axial stiffness A_{11},

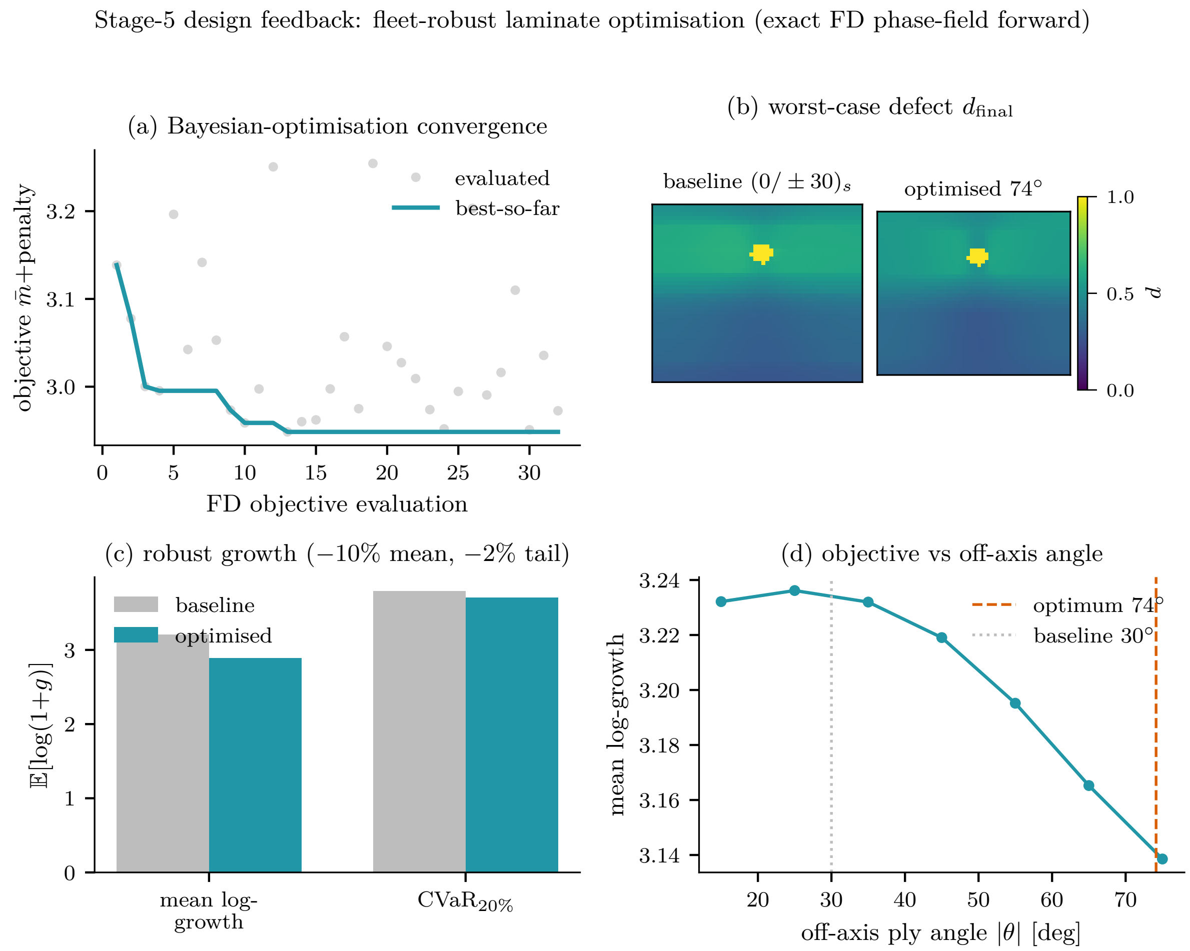

Figure 10: Stage 5. (a) BO convergence; (b) worst-case damage, baseline vs. optimised layup; (c) mean/CVaR growth; (d) objective vs. off-axis angle.

and bending/buckling D_{11} — restores an *interior* optimum: on the growth–stiffness Pareto front the knee sits at ~40°, and the optimal angle moves 75° → 15° as the stiffness weight grows (Fig. 11). The fleet defect distribution Stage 5 optimises against is drawn from the Stage-4 hierarchical posterior through a monotone growth-rate→severity link, rather than a synthetic prior — tightening the 4 → 5 handoff.

Robustness to the uncalibrated forward (honest limitation). The Stage-5 forward is the exact FD phase-field but with *representative* (uncalibrated) constants and a synthetic near-transition fleet load. We test how far the design *conclusion* survives this: perturbing the phase-field length scale ℓ and ply toughness G_c^{ply} by tens of percent and the fleet load by ±15%, then re-optimising (Fig. 12). The optimal off-axis angle is *not* robust — it spans 15°–75° across the perturbations — because phase-field growth is threshold-like: the uncalibrated constants set whether the operating point sits in the narrow sub-/super-critical *transition window* where the layup has leverage; outside it growth saturates (every design fails) or vanishes (every design is safe) and the optimum is ill-posed. The design-feedback *mechanism* (fleet-driven inverse design, the growth–stiffness trade-off, interface toughening as the dominant lever) is sound, but a *quoted optimal angle is not a determinate result* until the forward is calibrated — the motivation for the calibrated, layup-conditioned forward of the Keio continuation (§9).

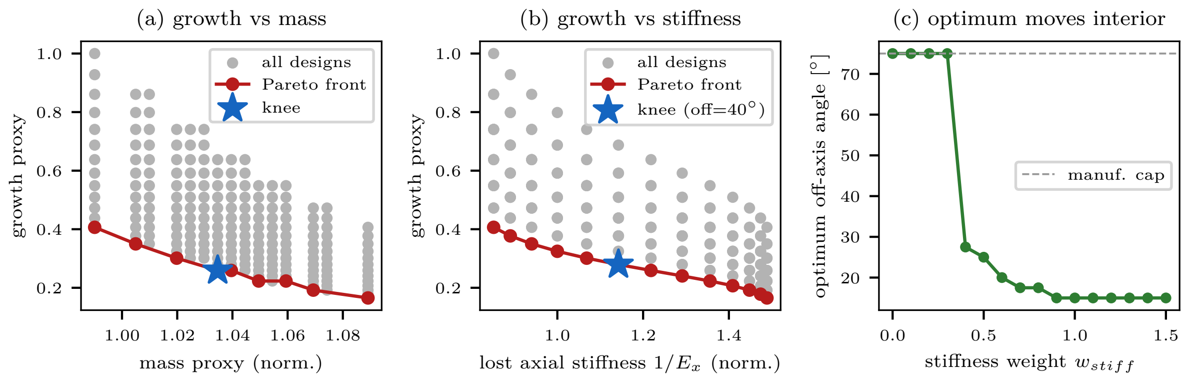

Figure 11: Stage 5, multi-objective. Growth vs. (a) mass and (b) axial-stiffness Pareto fronts with the utopia-distance knee; (c) the optimal off-axis angle moves off the manufacturable cap into the interior as the stiffness weight grows.

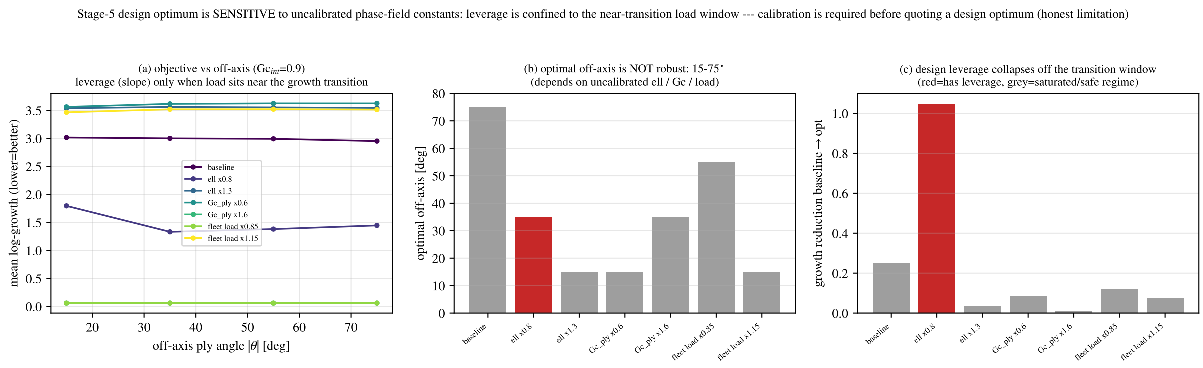

Figure 12: Stage 5 robustness. Perturbing the uncalibrated phase-field constants (ℓ, G_c^{ply}) and the fleet load: (a) the objective has off-axis leverage only near the growth transition; (b) the optimal off-axis angle spans 15°–75° (not robust); (c) the design improvement collapses once the constants push the operating point off the transition window.

9 Sim-to-real, domain adaptation, and the inverse-problem bridge

The stack is trained on synthetic FEM, so deployment must cross two distinct distribution gaps: a *data* gap (synthetic → measured fields) and an *inference* gap (in-distribution → out-of-distribution defects). This section closes both, and the second is the concrete bridge to the Keio continuation.

Real-data coverage — how far real measurements carry the pipeline. Before quoting any synthetic-grounded number we state, per stage, exactly how far *measured* data reaches. On the fairing — the structure with the most real measurements — we trace the chain end-to-end on real Open-Guided-Waves data (Fig. 13). *Stage 0 detection* runs on the real long-term guided-wave campaign (34 months, large-N healthy/damaged); it is genuinely temperature-confounded (mean FPR ~0.5 uncompensated) and is recovered only by the domain adaptation below — so detection is real but only deployable *with* the temperature correction. *Stage 1 classification* and *Stage 2 characterisation* run on the real stringer wavefield (Intact / 1st / 2nd impact): the per-sensor damage index is monotone in severity (mean DI $0 \to 0.57 \to 1.00$) and spatially localises the impact (peak-DI sensor moves $250 \to 467$ mm) — real signals, though with one measurement per state these are demonstrated as a damage-index pattern, not a large-N ML score. *Stage 3 prognosis onward is synthetic*: no real cycle/growth test exists for this article, so the clearance, fleet and design results below use the representative phase-field forward and a synthetic fleet. The honest headline is that detection/classification/characterisation are real-data-grounded and the *life*-side

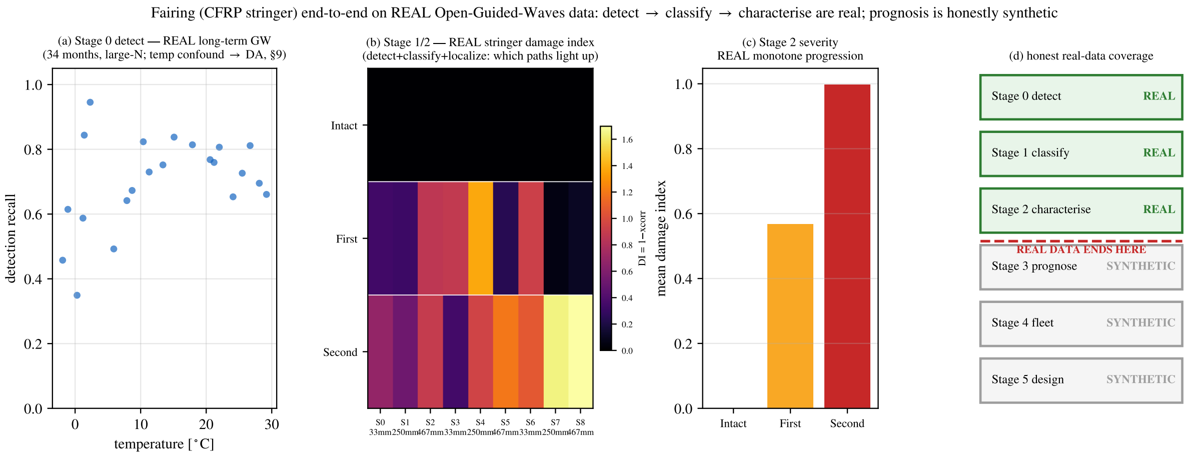

Figure 13: Real-data coverage on the fairing. (a) Stage-0 detection on the real long-term guided-wave set (recall vs. temperature — the confound DA must fix); (b) Stage-1/2 on the real stringer wavefield (per-sensor damage index: detect, classify 1st vs 2nd impact, and localise); (c) real monotone severity $0 \to 0.57 \to 1.00$; (d) the honest coverage map — detect/classify/ characterise are real, prognosis onward is synthetic.

of the stack is not yet — the single largest gap, and the reason the Kojima TSA full-field campaign (CFRP prosthetic specimens) and a real fatigue dataset are the priority next inputs.

Sim-to-real domain adaptation, on both real-data structures. The cross-structure real-data result is that an *unsupervised* per-feature quantile matching [26] aligns measured to FEM statistics where naive normalisation fails. For the *surface-stress modality* (interstage application, validated on prosthetic specimens), the stack ingests measured TSA-derived full-field DSPSS from a 22-specimen CFRP *prosthetic* campaign [2] and the quantile step closes ~97% of the FEM→measured gap (proxy-$\mathcal{A}$ distance $2.0 \to 0.3$, Fig. 15). On the *fairing*, the gap is a real long-term operational shift — guided-wave records collected across seasons carry a temperature drift — and we use it to stress-test the framework's distribution-free clearance guarantee. A split-conformal false-alarm threshold calibrated on spring-healthy data holds in distribution (empirical FPR $\approx \alpha$), but the seasonal shift breaks exchangeability and the guarantee collapses: at a nominal 5%, the *winter* false-alarm rate rises to ~68% (Fig. 14). Two corrections restore control to $\approx \alpha$ across seasons — the unsupervised per-feature quantile DA (FPR 14% $\to$ 5%, whereas z-score standardisation makes it *worse*, $\to$ 30%) and a temperature-conditional (Mondrian) conformal threshold [22]. Honestly, though, under the strong winter shift restoring *false-alarm* control by recalibration alone costs *detection* — cold-damage recall collapses — because the shift degrades the detector's *separability*, not only its calibration; recovering detection power needs feature-level DA, not just a recalibrated threshold. The OGW fairing is thus the framework's real-data confirmation that the conformal clearance guarantee is deployable only *with* an explicit environmental correction. The measured CFRP prosthetic TSA data are used under co-authorship with the campaign's owner.

Forward physics generalises where amortised inference does not. The inference gap motivates the converse of the amortised Stage-2 posterior, which never sees the forward physics. We prototype a JAX-differentiable anisotropic AT2 forward $F(\theta)$ (gradient finite-difference-verified to relative 3×10^{-7}) whose Jacobian supplies a physics-grounded Laplace posterior $\Sigma = \sigma^2 (J^\top J + \Lambda_{\text{prior}})^{-1}$ — no labelled (θ, x) pairs. Out of the training distribution the amortised posterior degrades ~25-fold and its credible intervals collapse to zero coverage, while the differentiable-physics posterior is essentially invariant (Table 4): *forward physics generalises; amortised inference extrapolates poorly*. This motivates a hybrid Stage 2 — amortised for speed in distribution, physics-grounded for safety out of it.

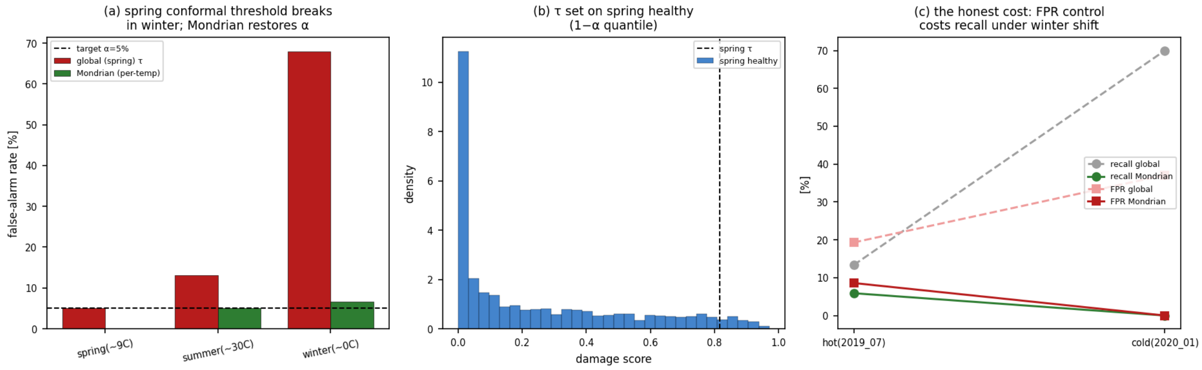

Figure 14: Conformal clearance guarantee under the real seasonal shift (fairing, OGW long-term). (a) a spring-calibrated split-conformal false-alarm threshold holds in distribution but the winter shift inflates the false-alarm rate to ~68% at a nominal 5%; a temperature-conditional (Mondrian) threshold restores $\approx \alpha$ across seasons. (b) the spring healthy-score calibration. (c) the honest cost: under the strong winter shift, restoring false-alarm control by recalibration alone collapses cold-damage recall — the shift degrades separability, so feature-level domain adaptation, not recalibration, recovers detection power.

Table 4: Amortised (FMPE) vs. differentiable-physics posterior, in- and out-of-distribution.

set	method	c_x RMSE	size RMSE	90% coverage
in-dist	FMPE	0.006	0.002	0.958
in-dist	diff-phys	0.001	0.003	1.000
OOD	FMPE	**0.144**	**0.096**	**0.000**
OOD	diff-phys	0.002	0.001	0.958

Fast differentiable forward. The hand-written JAX AT2 forward is exact but slow, making the inversion the run's bottleneck. Substituting the Stage-3 FNO surrogate (Fig. 8) as the differentiable forward keeps gradients exact — autograd matches finite differences to relative 7×10^{-4} on the decision heads — and collapses a MAP+Laplace inversion from an estimated ~530 s of FD Jacobian evaluations to 12 s (~43×). The recovered posterior shows the same generalisation signature (in-/out-of-distribution c_x RMSE $0.15 \to 0.39$, coverage $0.72 \to 0.56$), at a calibration cost — in-distribution coverage 0.72 rather than the exact forward's ~0.95 — that is the honest price of a smooth surrogate. This makes amortised-quality *differentiable* inversion interactive.

The bridge to the Keio continuation. These two threads converge on the same forward programme. The differentiable, fast forward is exactly what a TMCMC/Bayesian-inverse phase needs; and its physics — anisotropic phase-field fracture seeded by interlaminar cohesive zones — is the *prognosis* layer that the new torsion drive shaft (§2) extends from mode-I delamination to mode-II/III interlaminar shear. The CZM→phase-field prognosis demonstrated here on four structures, now under torsion, is therefore not only this paper's Stage 3: it is the computational-solid-mechanics foundation of the Keio phase-field × GNN-prognosis continuation, where a differentiable forward and fleet-scale Bayesian inversion close the loop on measured data. As a concrete instance, the torsion drive shaft's interlaminar debond grows by an unstable snap: across three viscous-regularisation levels an implicit static solve stalls at the same onset, whereas an explicit quasi-static solve (kinetic/internal energy 1–4% through the loading ramp) captures the full $0 \to 100\%$ debond, with the torque peaking at the onset (~$1.7°$ twist) and dropping post-peak as the interface fails (Fig. 16).

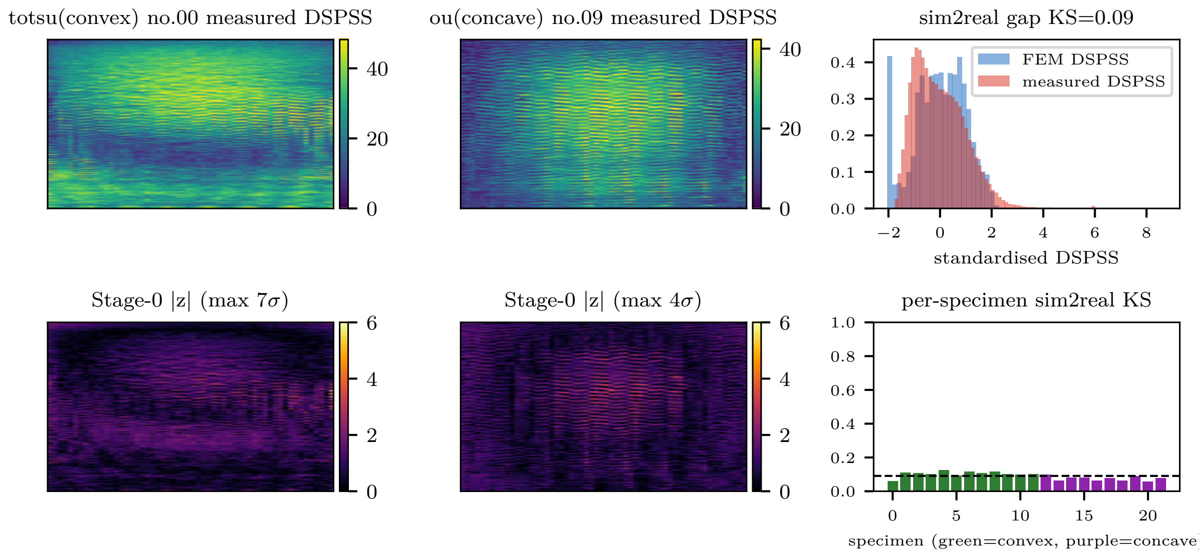

Figure 15: Sim-to-real of the surface-stress modality on real CFRP prosthetic specimens. Measured TSA-derived DSPSS vs. FEM, and the proxy-$\mathcal{A}$ distance before/after unsupervised quantile domain adaptation (2.0 → 0.3, ~97% of the gap closed).

10 End-to-end decision reliability

Every stage above is calibrated in isolation; an operator, however, only sees the *final* call. We therefore propagate uncertainty all the way to the clearance decision and ask how often it is right, using the exact FD phase-field as ground-truth physics so that no new measured data is required. For a known defect θ the next-flight growth is deterministic at fixed load, so the genuine aleatoric source is operational peak-load scatter: integrating the FD forward over that load distribution yields a true growth probability and, through the same expected-cost rule, an *oracle* decision. Four decisions then isolate each error contribution — oracle (θ_{true}+FD), surrogate-oracle (θ_{true}+FNO), FD-posterior (real FMPE θ+FD), and the shipped pipeline (FMPE θ+FNO) — over 40 held-out test scenarios spanning all three classes.

The end-to-end decision accuracy (pipeline vs. oracle) is **85**%, and the safety-critical **dangerous-miss rate** P(**OK** | **oracle=RETIRE**) **is 0**%: all 16 retire-worthy vehicles are caught (Fig. 17a), and every error is confined to the low-consequence OK↔REPAIR band. These figures are measured against a *synthetic* oracle — the exact FD phase-field integrated over the load distribution — because no measured clearance ground truth exists; they quantify internal error propagation (posterior + surrogate vs. exact physics), *not* agreement with real life-test outcomes, which remains future work (§9). The decomposition is informative: the Stage-2 posterior is the dominant error source (decision-flip rate 25%) over the surrogate forward (17.5%), and the two partially cancel to a 15% total (Fig. 17b) — a slightly conservative surrogate offsetting a slightly under-calling posterior. Finally, bootstrapping the posterior×load draws gives a per-decision confidence ("this OK call is 87% reliable"); validated against the oracle it is *overconfident* (ECE 0.18), since sampling spread alone misses the surrogate's systematic bias. A 1-D Platt rescaling of the confidence against decision-correctness, reported under leave-one-out cross-validation, removes most of it — ECE 0.18 → 0.06 and mean claimed reliability 96.5% → 84.5% against an empirical 85% (Fig. 17d); the rescaling is monotone, so the decision is unchanged and only its reliability estimate is corrected. The α, β thresholds, assumed at a 100:1 loss-to-repair ratio, we anchor to representative reusable-launch economics (refurb/loss 1/100–1/30, retire/loss 0.1–0.4): a Monte-Carlo over this range gives $\alpha = 0.035\ [0.021, 0.055]$ and $\beta = 0.53\ [0.28, 0.93]$, and shows that while an OK/REPAIR call near P_{grow}=0.1 is economically robust, the REPAIR-vs-RETIRE verdict at P_{grow}≈0.3–0.5 flips across plausible economics — a fragility invisible to a single assumed cost point.

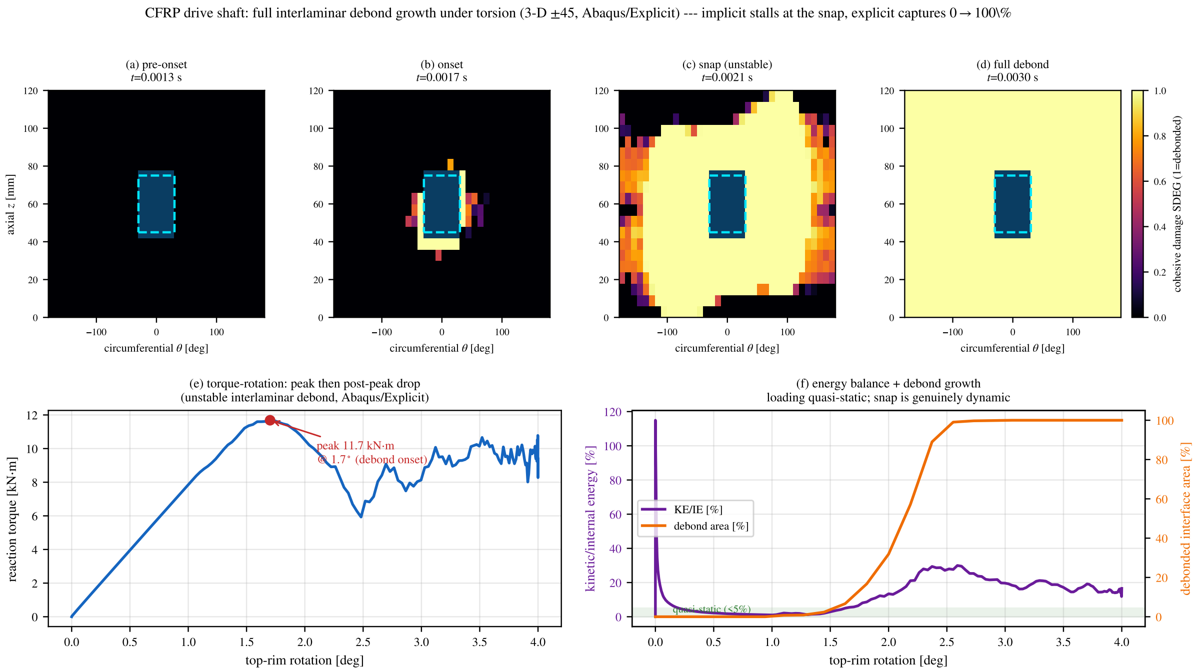

Figure 16: Full interlaminar debond growth in the torsion drive shaft (3-D ±45, Abaqus/Explicit). Top: cohesive damage (SDEG) on the unrolled θ–z interface at four times — pre-onset, onset, unstable snap, and full debond (dashed box = initial debond). Bottom: (e) torque–rotation with the post-peak drop; (f) kinetic/internal energy stays quasi-static ($< 5\%$) through loading while the debonded area runs $0 \longrightarrow 100\%$ — the snap is genuinely dynamic, which is why the implicit static solve cannot pass it.

11 Solver verification and detector robustness

Verification. The FD phase-field operator passes a method-of-manufactured-solutions test at second order in 2-D (interior order 2.04) and 3-D (order 2.05, a small-grid proof the scheme extends to volumetric delamination); a mesh-convergence study places the production n_x=60 grid at ∼0.4% error in peak reaction force (Fig. 18). Replacing the constant-peak load proxy with a variable-amplitude flight spectrum (liftoff / max-Q / buffet / MECO / re-entry / landing) run through the Carrara fatigue shows the proxy *under*-predicts mean life ∼3.4× by applying the rare max-Q amplitude to every cycle; anchoring the fatigue threshold $\bar{\alpha}_T$ to a representative CFRP S–N slope restores a well-defined Wöhler curve where the library default gives none.

Detector robustness and baselines. Stage-0 detection holds to AUROC 0.84 at sensor noise σ=0.1 of the field std — refuting a folklore "collapse at 0.1" — and only reaches chance near $\sigma\approx$0.7–1.0, with lock-in frame averaging ($\sigma/\sqrt{N}$) recovering AUROC ≥ 0.8 in 32 frames; the reference-free $|z|$ detector needs no healthy twin (the new-build first-flight case). Benchmarked against standard anomaly detectors (PCA-reconstruction, isolation forest, one-class SVM, LOF) on labelled coupons, the physics $|z|$ ties the best on AUROC and leads on AUPRC for the rare (<1%) defect class — the defect is a stress concentration, so its magnitude *is* the signal, and a trivial physics score matches tuned outlier detectors.

12 Discussion

Several findings cut across the stack. On fixed geometry a simple per-node statistic dominates the learned anomaly detectors (Table 2): when the healthy manifold is a tight cloud and the *same* structure is re-inspected every turnaround, distance-to-normal is hard to beat, and the learned models earn their place not at detection but at the semantic label (type/region/depth) and at transfer to new airframes and repairs.

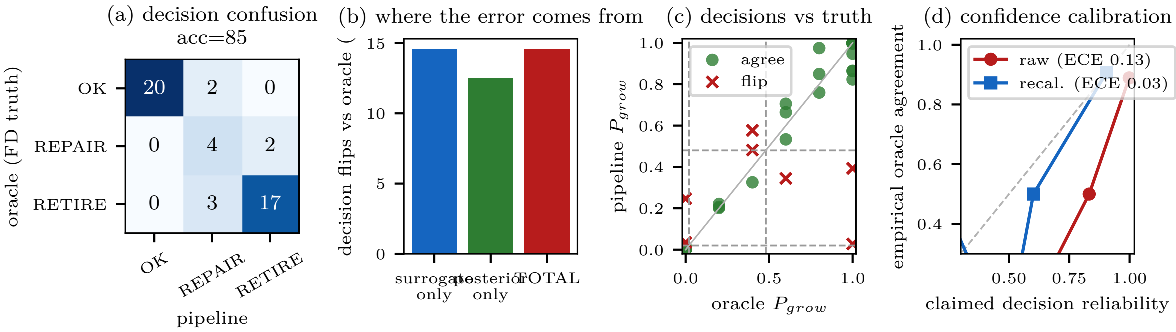

Figure 17: End-to-end decision UQ. (a) pipeline-vs-oracle decision confusion (RETIRE row clean); (b) error decomposition (posterior > surrogate, partial cancellation); (c) pipeline vs. oracle P_{grow} with the α, β thresholds, flips at the boundaries; (d) per-decision confidence calibration.

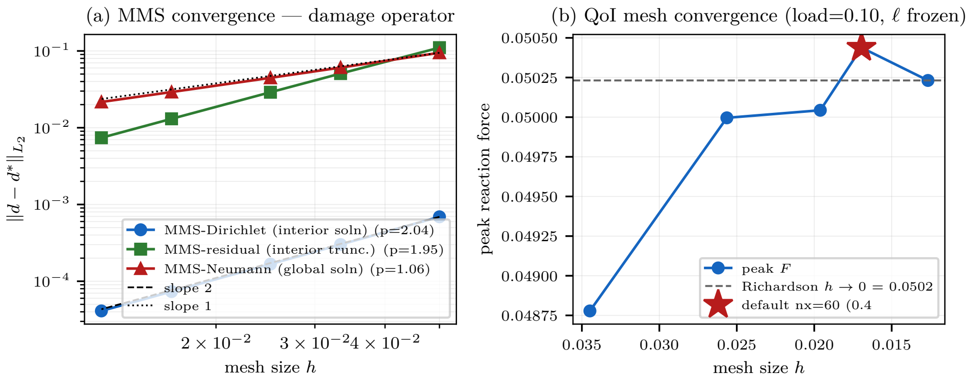

Figure 18: Solver verification. (a) MMS log–log error vs. mesh size with fitted slopes (interior 2nd-order; the zero-flux Neumann closure is 1st-order, shown honestly); (b) peak reaction force converging to the Richardson asymptote, the production n_x=60 grid starred.

The decision core itself transfers across geometry and sensing modality but *not* across an unseen load mode: structure-agnosticism holds for the cost-optimal thresholds and confidence maps, while the growth physics must be re-anchored per load mode. Cast this way, the framework formalises the condition-based, staged-certification, fleet-leader practice of operational reuse and makes it usable — as a calibrated decision — from the first article, rather than only after a fleet history accrues.

The limitations are deliberate and stated throughout. The operational load is a normalised proxy; the fatigue degradation is not yet S–N-anchored; the AT2 strength is length-scale-dependent (trends, not absolute values); and defect depth is only weakly observable from a surface field — a physics limit of thermoelastic measurement, not a model failure. Consequently the life side is only partly real-data-grounded: the one calibrated life-side quantity, the Paris exponent from real DCB data, already flips a mid-size clearance, which both validates the seeding pathway and signals how much a fully calibrated life side would move the decision. The framework's distribution-free guarantees are likewise necessary but not sufficient under real environmental shift — a season-calibrated conformal false-alarm guarantee collapses in winter and is restored only with explicit temperature compensation, and restoring false-alarm control alone does not recover the detection power the shift destroys. Closing the life-side real-data gap is therefore the priority, and it motivates the continuation: a differentiable, calibrated phase-field forward that supplies physics-grounded posteriors and gradients for the inverse problem, where amortised inference extrapolates poorly.

13 Conclusion

Towards measured data. The sim-to-real bridge is under way: the stack ingests measured TSA-derived full-field DSPSS from a 22-specimen CFRP prosthetic campaign, and an unsupervised domain-adaptation step (per-feature quantile matching) closes ~97% of the FEM→measured distribution gap (proxy-$\mathcal{A}$ distance $2.0 \rightarrow 0.3$); a labelled measured-data detection benchmark awaits per-specimen defect masks.

Reproducibility. All stages are implemented and tested (~250 unit tests) in two public-on-request repositories, including a one-command end-to-end run (detect→classify→characterise→prognose →clear→fleet-update in <4 s), a decision-level UQ harness that scores the final clearance call against the exact FD oracle, and verification (MMS / mesh-convergence, 2-D and 3-D), robustness (SNR sweep, baseline benchmark) and economic-sensitivity studies; figures are generated by the scripts under `paper_figs/`.

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